

Chef Classification from Recipe Text using DistilBERT

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October 6, 2025

1 Overview

We fine-tuned DistilBERT-base-uncased to assign each recipe in a 6-class corpus to its chef. After removing 14 exact duplicates, we retained 2,985 unique recipes and produced an 80/20 stratified split. The best checkpoint (early stopping after 5 epochs) reaches 92.13% validation accuracy and 0.916 macro-F1, a gain of roughly 49 percentage points over the strongest TF-IDF baseline.

2 Data Preparation

- Parsed list fields (tags, ingredients, steps) and concatenated five text sections in the order *name* $\rightarrow$ *ingredients* $\rightarrow$ *tags* $\rightarrow$ *description* $\rightarrow$ *steps*. This protects short, identifying fields from truncation.
- Dropped duplicated concatenated texts prior to splitting to prevent leakage (class counts range from 365 to 801 after deduplication).
- Tokenised with DistilBERT’s WordPiece tokenizer at max length 512. The deduplicated corpus has a median of 234 tokens and 95th percentile of 418, so truncation is rare.

3 Model and Training

- DistilBERT encoder with dropout 0.2, a $768 \rightarrow 768$ projection, ReLU activation, and a final linear layer producing six logits.
- Optimiser: AdamW with learning rate 2×10^{-5} , weight decay 0.01, linear warmup across 6% of steps, gradient clipping at 1.0.
- Chill-mode batches (8 for training, 16 for evaluation) on Apple MPS. Evaluation every 100 steps with patience two evaluations keeps the best macro-F1 checkpoint.

4 Results

Table 1: Validation accuracy and macro-F1.

Method	Accuracy	Macro-F1
TF-IDF (description)	30.0%	—
TF-IDF (all fields)	43.0%	—
DistilBERT (best checkpoint)	92.13%	0.916

Key observations:

- Validation accuracy and macro-F1 peak between epochs 4 and 5; later checkpoints oscillate slightly but do not improve macro-F1.

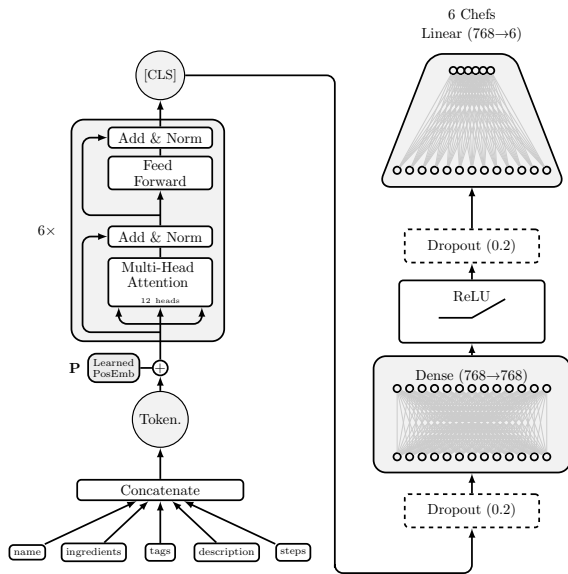


Figure 1: Text-only DistilBERT pipeline used for chef classification.

- Prediction distribution on the blind test set stays within ± 2.3 percentage points of the training prior (Chef 1533 mildly over-predicted, Chef 3288 slightly under-predicted).
- Misclassifications mostly confuse chefs 1533 and 3288, whose recipes overlap on small appetisers versus make-ahead meals.

5 Lessons and Next Steps

- Deduplicating texts before splitting was essential; earlier runs that skipped this step overstated validation accuracy.
- The straightforward DistilBERT head is sufficient once evaluation cadence and early stopping are tightened; longer schedules offer little extra gain.
- Future improvements could add non-textual

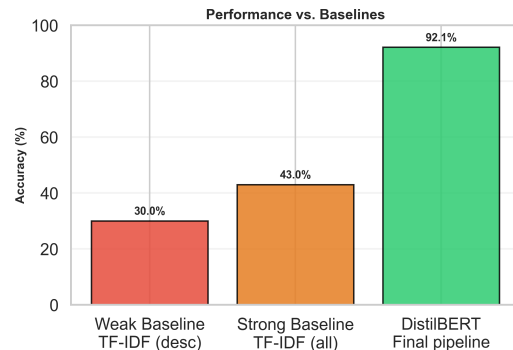


Figure 2: Accuracy improvement over TF-IDF baselines.

cues (quantities, durations), explore cross-validation for more stable estimates, and analyse attention patterns to separate stylistic cues from obvious keywords.

Bibliography

References